Principal Component Analysis [PCA]

Concept:

To transform a given set of features in to new reduced set without any loss of information.

Consider a vector of the form,

$$x = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$$

The mean vector of x is defined as

$$m_x = E\{x\}$$

i.e $$m_x = \frac{1}{M} \sum_{k=1}^{M} x_k$$

From this, the covariance matrix can be calculated as

$$C = E\{(x - m_x)(x - m_x)^T\}$$

i.e $$A = \frac{1}{M} \sum_{k=1}^{M} x_k x_k^T - m_x m_x^T$$

Here A is called transform kernel.

The mapping of vector x to y is defined as

$$y = A(x - m_x)$$

This transform is called Karhunen-Loeve or Hoteling transform.

The original vector x can be reconstructed by

$$x = A^T y + m_x$$

PCA Algorithm steps

1. The target dataset x is obtained
2. The mean is subtracted from the dataset. Let the mean be m. The adjusted dataset is $x-m$. The objective of this process is to transform the dataset with zero mean.
3. The covariance of dataset x is obtained. Let it be C
4. Eigen values and eigen vectors of the covariance matrix are calculated.
5. The eigen vector of the highest eigen value is the principal component of the dataset. The eigen values are arranged in descending order. The feature vector is formed with these eigen vectors

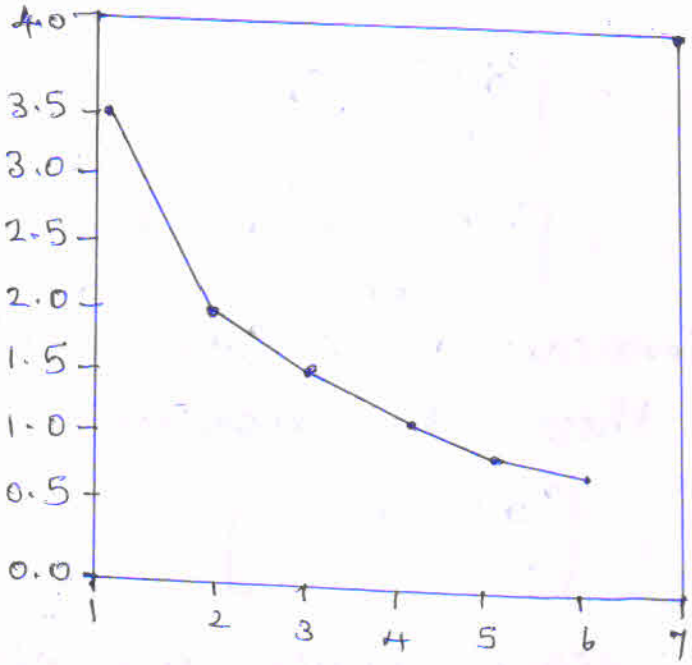
$$\text{Feature vector} = \{\text{eigen vector}_1, \text{eigen vector}_2 \ldots \text{eigen vector}_n\}$$

6. Obtain the transpose of feature vector. Let it be A.
7. PCA transform is $y = A \times (x-m)$ where x is input dataset, m is the mean, A is transpose of feature vector.
8. Original data can be retrieved using,

$$\text{original data}(f) = \{(A)^{-1} * y\} + m$$

$$= \{(A)^T * y\} + m$$

* The new data is dimensionally reduced matrix that represents the original data
* PCA is effective in removing the attributes that do not contribute.
* If the original data is obtained from new data, it is called as lossless.
* Scree plot is a visualization technique to visualize the principal components or variables that play more important role.
* Example : For the dataset with 246 attributes, PCA is applied & its scree plot is shown below.



Scree plot indicates,

* Only 6 out of 246 attributes are important
* First attribute is more important than all other attributes.

Sample Problem:

Let the data points be $\begin{bmatrix} 2 \\ 6 \end{bmatrix}$ and $\begin{bmatrix} 1 \\ 7 \end{bmatrix}$. Apply PCA and find the transformed data. Again apply the inverse and prove that PCA works.

Step1 : Combine 2 vectors into matrix & compute mean.

$$\text{Mean } \underset{\text{or } m}{\mu} = \begin{bmatrix} \frac{2+1}{2} \\ \frac{6+7}{2} \end{bmatrix} = \begin{bmatrix} \frac{3}{2} \\ \frac{13}{2} \end{bmatrix} = \begin{bmatrix} 1.5 \\ 6.5 \end{bmatrix}$$

Step2 : Get the adjusted data by subtracting mean from data.

$$x_1 = \begin{bmatrix} 2-1.5 \\ 6-6.5 \end{bmatrix} = \begin{bmatrix} 0.5 \\ -0.5 \end{bmatrix}$$

$$x_2 = \begin{bmatrix} 1-1.5 \\ 7-6.5 \end{bmatrix} = \begin{bmatrix} -0.5 \\ 0.5 \end{bmatrix}$$

Step3 : Obtain covariance of data vectors

$$m_1 = \begin{bmatrix} 0.5 \\ -0.5 \end{bmatrix} \begin{bmatrix} 0.5 & -0.5 \end{bmatrix} = \begin{bmatrix} 0.25 & -0.25 \\ -0.25 & 0.25 \end{bmatrix}$$

$$m_2 = \begin{bmatrix} -0.5 \\ 0.5 \end{bmatrix} \begin{bmatrix} -0.5 & 0.5 \end{bmatrix} = \begin{bmatrix} 0.25 & -0.25 \\ -0.25 & 0.25 \end{bmatrix}$$

Final covariance matrix C is obtained by adding these two matrices

$$C = \begin{bmatrix} 0.5 & -0.5 \\ -0.5 & 0.5 \end{bmatrix}$$

Step 4 : Obtain eigen values and eigen vectors of matrix C.

$$= \begin{bmatrix} 0.5-\lambda & -0.5 \\ -0.5 & 0.5-\lambda \end{bmatrix}$$

$$= \left[(0.5-\lambda)\times(0.5-\lambda)\right] - \left[(-0.5)\times(-0.5)\right] = 0$$

$$= (0.25 - 0.5\lambda - 0.5\lambda + \lambda^2) - (0.25) = 0$$

$$= \lambda^2 - \lambda + \cancel{0.25} - \cancel{0.25} = 0 \Rightarrow \lambda^2 - \lambda = 0$$

$$= \lambda(\lambda - 1) = 0$$

$$\Rightarrow \lambda = 0 \text{ (or) } \lambda = 1 \quad \text{[Eigen values]}$$

To obtain eigen vector,

$$(C - \lambda I) \times X = 0$$

For $\lambda = 0$, $\begin{bmatrix} 0.5 & -0.5 \\ -0.5 & 0.5 \end{bmatrix} - \left[0 \times \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}\right] \times \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$

$$0.5x_1 - 0.5x_2 = 0 \Rightarrow x_1 - x_2 = 0 \rightarrow (1)$$

$$-0.5x_1 + 0.5x_2 = 0 \Rightarrow -x_1 + x_2 = 0 \rightarrow (2)$$

By taking $x_1 - x_2 = 0$ from (1) we can get

$$\Rightarrow x_1 = x_2 \rightarrow (3)$$

Substitute (3) in (1)

$$x_2 - x_2 = 0$$

$$\Rightarrow x_2 = x_2 \rightarrow (4)$$

Let us assign $x_1 = x_2 = k$ with $k = 1$

$\Rightarrow$ Eigen vectors are $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$

For $\lambda = 1$, $\left[\begin{bmatrix} 0.5 & -0.5 \\ -0.5 & 0.5 \end{bmatrix} - 1 \times \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}\right] \times \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 0$

$$= \begin{bmatrix} 0.5 & -0.5 \\ -0.5 & 0.5 \end{bmatrix} - \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \times \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 0$$

$$= \begin{bmatrix} -0.5 & -0.5 \\ -0.5 & -0.5 \end{bmatrix} \times \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 0$$

$$\Rightarrow \left.\begin{array}{l} -0.5x_1 - 0.5x_2 = 0 \\ -0.5x_1 - 0.5x_2 = 0 \end{array}\right\} \Rightarrow -x_1 - x_2 = 0 \Rightarrow x_1 = -x_2$$

Substituting $-x_2$ in x_1

$$-x_2 - x_2 = 0 \Rightarrow x_2 = x_2$$

Let us assign $x_1 = x_2 = K$ with $K = 1$

Eigen vectors are $\begin{bmatrix} -1 \\ 1 \end{bmatrix}$

$\therefore$ Eigen vectors for $\lambda = 0$ & $\lambda = 1$ are

$$\begin{bmatrix} 1 \\ 1 \end{bmatrix} \& \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

After sorting obtain A as,

$$A = \begin{bmatrix} -1 & 1 \\ 1 & 1 \end{bmatrix}$$

$$A^T = \begin{bmatrix} -1 & 1 \\ 1 & 1 \end{bmatrix}$$

Steps : Normalize vector A by dividing each elements of the vector by norm of the vector

$$\therefore A = \begin{bmatrix} -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix}$$

Step 6 : Compute transformed matrix y as

$$y = A \times \underbrace{(x - m)}_{\text{adjusted matrix}}$$

$$= \begin{bmatrix} -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix} \begin{bmatrix} 0.5 & -0.5 \\ -0.5 & 0.5 \end{bmatrix}$$

$$= \begin{bmatrix} -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix} \begin{bmatrix} \frac{1}{2} & -\frac{1}{2} \\ -\frac{1}{2} & \frac{1}{2} \end{bmatrix}$$

$$= \begin{bmatrix} -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ 0 & 0 \end{bmatrix}$$

Step 7 : Obtain the original matrix by

$$x = (A)^T * y + m$$

$$x = A^T y + m = \begin{bmatrix} -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix} \begin{bmatrix} -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ 0 & 0 \end{bmatrix} + \begin{bmatrix} 1.5 \\ 6.5 \end{bmatrix}$$

$$= \begin{bmatrix} \frac{1}{2} & -\frac{1}{2} \\ -\frac{1}{2} & \frac{1}{2} \end{bmatrix} + \begin{bmatrix} 1.5 \\ 6.5 \end{bmatrix} = \begin{bmatrix} 2 & 1 \\ 6 & 7 \end{bmatrix}$$

$\therefore$ Original matrix is obtained without any loss of information.

Linear Discriminant Analysis [LDA]

* Feature reduction technique
* Used to project higher dimension data to lower dimension data
* Used to classify the data
* Let two classes C_1 & C_2
* Compute mean of C_1 & C_2 as

$$\mu_1 = \frac{1}{N_1} \sum_{x_i \in C_1}^{n} x_i \,, \quad \mu_2 = \frac{1}{N_2} \sum_{x_i \in C_2}^{n} x_i$$

* The aim of LDA is to optimize the function

$$J(V) = \frac{V^T \sigma_B V}{V^T \sigma_W V} \,,$$

V - Linear projection
σ_B - Class scatter matrix
σ_W - within scatter matrix

$$\sigma_B = N_1 (\mu_1 - \mu)(\mu_1 - \mu)^T + N_2 (\mu_2 - \mu)(\mu_2 - \mu)^T$$

$$\sigma_W = \sum_{x_i \in C_1} (x_i - \mu_1)(x_i - \mu_1)^T + \sum_{x_i \in C_2} (x_i - \mu_2)(x_i - \mu_2)^T$$

* The maximization of $J(V)$ should satisfy the eqn.

$$\sigma_B V = \lambda \sigma_B V \quad \text{or} \quad \sigma_W^{-1} \sigma_B V = \lambda V$$

As $\sigma_B V$ is always in the direction of $(\mu_1 - \mu_2)$, V can be given as

$$V = \sigma_W^{-1} (\mu_1 - \mu_2)$$

* Let $V = \{v_1, v_2, \ldots v_d\}$ be generalized eigen vectors of σ_B & σ_W, where d is largest eigen values
* The transformation of x is given as

$$y = V_d^T x$$